

Notes: George & Waldfogel (JPE, 2003)

Who Affects Whom in Daily Newspaper Markets?

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1 Big picture

- **Question.** Do consumers in differentiated-product markets benefit more from other consumers who share their preferences than from consumers with different preferences?
- **Core mechanism.** When fixed costs are large, firms cannot target every niche. Larger consumer groups attract products closer to their tastes. This creates a **preference externality**: one person's presence can improve product availability for others with similar preferences.
- **Setting.** U.S. daily newspaper markets are well suited for the test because daily newspapers have high fixed costs, markets support few products, newspaper content is positionable, and media preferences differ sharply by race and Hispanic status.
- **Data.** The main data combine zip-code-level newspaper circulation, zip-code demographics, MSA demographics, and newspaper staffing/content measures. The final circulation-demographics sample has 11,584 zip codes in 269 newspaper markets.
- **Empirical object.** The authors ask whether the tendency of group members to buy a newspaper rises more with their own group's market population than with the population of other groups.
- **Main race result.** Black newspaper readership rises with the number of blacks in the market and falls with the number of whites. White readership rises modestly with the number of whites and is mostly unaffected by the number of blacks.
- **Hispanic result.** Similar patterns appear for Hispanics and non-Hispanics, especially in MSA fixed-effect specifications.
- **Other demographic groups.** The evidence is weak for groups defined by education, income, or age, partly because those groups are less discrete, less sharply separated in media tastes, or less variable across markets.
- **Mechanism evidence.** Product positioning appears to matter. In cities with competing broadsheets and tabloids, newspaper choice varies systematically with zip-code demographics. Across MSAs, the content mix of newspapers varies with the fraction black.
- **Economic takeaway.** In differentiated-product markets, the welfare of a consumer group can depend on the size of its own group because product supply and product positioning respond to local demand composition.

2 Incremental contribution of the paper

- **From demand estimation to supply-side product targeting.** Prior differentiated-product work often takes product availability as given. This paper studies how the composition of consumers in a market affects the products that firms supply or the way firms position existing products.
- **Empirical evidence on preference externalities.** The paper brings the idea of preference externalities into an observable market setting: if more consumers of type g are present, firms target type g more strongly, changing consumption probabilities for both type g and other types.
- **A method for recovering group-specific effects without group-specific sales.** The authors do not observe newspaper sales separately by race within each zip code. They use cross-zip-code variation in demographic shares within MSAs to infer group readership gaps and then relate those gaps to MSA group populations.
- **Within-group and cross-group effects.** The paper separates own-group effects from cross-group effects. This distinction is central because market size can increase product variety for everyone, while product repositioning toward one group can make the product less attractive to other groups.

- **Mechanism test using product positioning.** The paper goes beyond circulation outcomes by using staffing categories and two-newspaper cities to show that different demographic groups prefer different content mixes and that content supply varies with market demographics.
- **Media-market relevance.** The paper connects industrial organization of differentiated products to media economics and political information. Newspapers are not just consumer goods; their availability and targeting may shape access to local information.
- **Welfare implication.** The paper suggests that the distribution of product-market welfare depends on group size. Minority consumers may receive worse product matches in markets where their group is small, even at similar prices.

3 Data and measurement

3.1 Market, circulation, and demographic data

- The unit of observation in the main analysis is the zip code.
- Newspaper circulation comes from Audit Bureau of Circulations data for roughly 1,200 daily newspapers.
- Zip-code circulation is averaged over 1996, 1997, and 1998, then summed across daily newspapers to form zip-code total daily circulation.
- The market is the MSA of publication. The working sample excludes nonmetropolitan zip codes, zip codes not matched to the 1990 Census, and MSAs with incomplete circulation data.
- Demographic information comes from the 1990 Census and includes race, Hispanic status, education, income, and age.
- The main readership outcome is circulation per capita, not circulation per household, because per capita circulation links directly to demographic shares.

Interpretation: The paper studies how market-level consumer composition affects the products that all zip codes in that market face. Therefore, it combines zip-code variation in local demographics with MSA-level variation in group populations.

3.2 Product positioning and content data

- The paper uses Burrelle’s Media Directory to count reporters and editors by subject specialty.
- Hard-news categories include news, business, government, and related topics.
- For each MSA, the authors compute the fraction of reporters/editors assigned to hard-news topics across newspapers in the market.
- In four two-newspaper markets, the authors compare an upscale broadsheet with a tabloid or less hard-news-oriented competitor.

Interpretation: These content measures are used to test whether product positioning is a plausible mechanism behind the preference externalities in circulation.

3.3 Group definitions

- The strongest tests use race: black versus white.
- A second set of tests uses Hispanic status: Hispanic versus non-Hispanic.

- Additional tests use education, income, and age: college educated versus noncollege; poor versus nonpoor; age 65+ versus younger.
- Race and Hispanic status are expected to be the best contexts because preferences differ strongly and market shares vary substantially across MSAs.

Interpretation: Continuous or broad demographic dimensions may mix people with similar tastes into both groups. This makes preference externalities harder to detect for education, income, and age.

4 Models and empirical specifications

4.1 Simple model of group-specific readership

If group-specific newspaper sales were observed, the natural equations would be

$$s_z^W = \alpha_0 + \alpha_1 W_M + \alpha_2 B_M + e_z^W, \quad (1)$$

$$s_z^B = \beta_0 + \beta_1 W_M + \beta_2 B_M + e_z^B, \quad (2)$$

where s_z^W and s_z^B are white and black newspaper purchase rates in zip code z , W_M is the white population in market M , and B_M is the black population.

Interpretation: α_1 is the effect of more whites on white readership, β_2 is the effect of more blacks on black readership, and α_2, β_1 are cross-group effects. A preference externality predicts that own-group effects should exceed cross-group effects.

4.2 Recovering group effects without group-specific sales

The authors observe total per capita newspaper sales s_z and the zip-code black share b_z , not sales by race. Overall sales are

$$s_z = b_z s_z^B + (1 - b_z) s_z^W. \quad (3)$$

Substituting the group-specific equations gives

$$s_z = \alpha_0 + \alpha_1 W_M + \alpha_2 B_M + (\beta_0 - \alpha_0) b_z \quad (4)$$

$$+ (\beta_1 - \alpha_1) W_M b_z + (\beta_2 - \alpha_2) B_M b_z + \nu_z. \quad (5)$$

Interpretation: Cross-zip-code variation in b_z identifies the black-white readership gap within an MSA. Interacting b_z with MSA group populations shows how that gap changes when the market has more blacks or whites.

4.3 Interpreting the interaction terms

- If $(\beta_1 - \alpha_1) < 0$, then a larger white population raises white readership more than black readership; the black-white readership gap widens.
- If $(\beta_2 - \alpha_2) > 0$, then a larger black population raises black readership more than white readership; the black-white readership gap narrows.
- In MSA fixed-effect specifications, the levels $\alpha_1, \alpha_2, \beta_1, \beta_2$ are not separately identified, but the gap terms remain identifiable.

Interpretation: The design is not a randomized experiment. Its strength is that it exploits within-market demographic variation to infer group-specific readership gaps, then compares these gaps across markets with different group sizes.

4.4 Weighting and standard errors

- Regressions are weighted by zip-code population.
- Standard errors are clustered by MSA because the key regressors are at the MSA level.
- Specifications add census region controls, region controls interacted with group share, MSA observables, and MSA fixed effects.

Interpretation: Clustering addresses correlated errors within markets. Fixed effects address market-level unobservables but sacrifice identification of the direct MSA population terms.

5 Identification and empirical design

5.1 What identifies preference externalities

- Within an MSA, zip codes differ in demographic composition and circulation per capita.
- This variation identifies a group readership gap for each market.
- Across MSAs, the size of each demographic group varies.
- If the group readership gap changes systematically with group population, this is interpreted as evidence that market composition affects product targeting and consumption.

Interpretation: The key identifying structure is a two-level comparison: within-market demographic variation reveals the group gap; across-market population variation explains the gap.

5.2 Why newspapers are a useful setting

- Daily newspapers have large fixed costs, so markets support limited product variety.
- Newspapers can change content and positioning without necessarily changing prices.
- Media preferences vary sharply by race and Hispanic status.
- Newspaper consumption has outside substitutes, so consumers may abstain if the local newspaper is poorly matched to their tastes.

Interpretation: The market has the core ingredients for preference externalities: heterogeneous tastes, fixed costs, limited variety, and market-level variation in group composition.

5.3 Main empirical caveats

- MSA demographics may be correlated with unobserved market traits that affect readership.
- Zip-code circulation does not reveal individual-level purchases by race or ethnicity.
- Newspaper content is measured imperfectly by reporter/editor categories.

- Product positioning is shown through associations, not through an experimental shift in newspaper content.
- The paper abstracts from price variation because newspaper subscription prices vary little across markets.

Interpretation: The evidence is strongest as a coherent pattern: readership results by group align with content-positioning evidence, and both align with differentiated-product theory.

6 Tables and figures

6.1 Table 1: Summary Statistics

Read:

- The main sample contains 11,584 zip codes and 269 MSAs.
- Average zip-code circulation per capita is 0.205; average MSA circulation per capita is 0.234.
- The MSA fraction black averages 0.100, with a 5th–95th percentile range of 0.008 to 0.290.
- The MSA fraction Hispanic averages 0.069, with a 5th–95th percentile range of 0.004 to 0.242.
- Race and Hispanic shares vary much more across markets than education, income, and age shares, making them better dimensions for testing market-composition effects.
- The average MSA fraction of reporters/editors in hard news is 0.585.

Economic message: The data contain substantial cross-market variation in race and Hispanic composition, exactly the variation needed to test whether consumer groups affect product supply and consumption.

TABLE 1
SUMMARY STATISTICS

		PERCENTILE				
	MEAN	5th	25th	50th	75th	95th
A. Population Statistics (1990 Census)						
Zip codes (N=11,584):						
Total population	13,795	590	2,827	8,639	21,433	41,663
Fraction black	.086	.000	.002	.014	.079	.479
Fraction Hispanic	.055	.000	.004	.012	.041	.266
Fraction college educated	.187	.045	.092	.147	.244	.471
Fraction poor (income <\$15,000)*	.228	.062	.136	.208	.299	.463
Fraction old (age ≥65)	.122	.042	.085	.116	.150	.217
MSAs (N=269):						
Total population (millions)	.574	.119	.293	.907	2.444	6.253
Black population (millions)	.072	.001	.016	.088	.217	1.085
Fraction black	.100	.008	.046	.083	.176	.290
Hispanic population (millions)	.046	.001	.006	.019	.136	.870
Fraction Hispanic	.069	.004	.010	.027	.065	.242
College-educated population	.081	.011	.036	.130	.344	1.207
Fraction college educated	.122	.081	.114	.132	.152	.207
Poor population (income <\$15,000)	.129	.030	.074	.200	.444	1.149
Fraction poor (income <\$15,000)	.258	.151	.195	.223	.276	.328
Old population (age ≥65)	.069	.014	.038	.105	.273	.784
Fraction old (age ≥65)	.124	.080	.106	.119	.137	.174
B. Circulation Statistics (Audit Bureau of Circulations)						
Circulation per capita:						
Zip codes (N=11,584)	.205	.020	.122	.192	.261	.431
MSAs (N=269)	.234	.149	.202	.233	.264	.318
C. Newspaper Statistics (<i>Burrelle's Media Directory</i>)						
MSAs: fraction hard news (N=269)	.585	.407	.530	.580	.640	.750

* Number of observations for this variable is 11,576.

Table 1: Summary Statistics

6.2 Table 2: Regressions of Newspaper Sales on Population Composition by Race and Table 3: Regressions by Hispanic Status

Read:

- Table 2 shows strong race-based preference externalities. The zip-code black fraction is negative, indicating lower baseline newspaper purchase rates among blacks than whites.
- In Table 2, the implied within-black effect β_2 is positive and statistically significant in the main specifications. Increasing the black population by one million raises black readership substantially relative to its low base.
- The implied cross effect of whites on black readership, β_1 , is negative: larger white populations make newspapers less attractive to black consumers relative to white consumers.
- The MSA fixed-effect specification confirms the key gap evidence: the black-white readership gap widens with white population and narrows with black population.
- Table 3 repeats the design for Hispanics and non-Hispanics. The basic specifications are weaker, but the MSA fixed-effect specification shows a pattern similar to race: Hispanic population raises Hispanic readership relative to non-Hispanic readership, while non-Hispanic population lowers Hispanic readership relative to non-Hispanic readership.

Economic message: These two tables are the core evidence. Consumers benefit more from same-type market size than from different-type market size, consistent with product targeting under fixed costs.

TABLE 2
REGRESSIONS OF NEWSPAPER SALES ON POPULATION COMPOSITION BY RACE (N=11,584)
Dependent Variable: Zip Code per Capita Newspaper Sales, Blacks and Whites

	(1)	(2)	(3)	(4)	(5)
MSA white population (α_1)	.0081 (1.24)	.0078 (1.94)*	.0078 (2.11)*	.0012 (.35)	
MSA black population (α_2)	.0016 (.06)	-.0002 (.01)	-.0005 (.02)	.0230 (1.25)	
Zip black fraction	-.1419 (13.58)**	-.1183 (9.79)**	-.1952 (5.88)**	-.2214 (7.56)**	-.1271 (12.57)**
Zip black fraction $\times$ MSA white population ($\beta_1 - \alpha_1$)	-.0440 (2.52)*	-.0430 (2.86)**	-.0412 (2.86)**	-.0287 (2.40)*	-.0290 (2.41)*
Zip black fraction $\times$ MSA black population ($\beta_2 - \alpha_2$)	.1598 (2.28)*	.1310 (2.24)*	.1295 (2.27)*	.0768 (1.61)	.0784 (1.60)
β_1	.161 (2.77)**	.131 (2.55)*	.129 (2.29)*	.100 (2.51)*	
β_2	.161 (3.02)**	.131 (2.42)*	.129 (2.18)*	.100 (2.24)*	
MSA fraction Hispanic				-.0617 (1.73)	
MSA fraction college educated				.3816 (3.75)**	
MSA fraction poor (income <\$15,000)				.0730 (.99)	
MSA fraction old (age ≥ 65)				.3474 (3.57)**	
Census regions	no	yes	yes	yes	no
Interacted regions	no	no	yes	yes	no
MSA fixed effects	no	no	no	no	yes
Constant	.2112 (38.85)**	.1796 (9.49)**	.1764 (8.08)**	.0697 (1.48)	.2259 (233.38)**

NOTE.—All regressions are population weighted with standard errors clustered by MSA. Constants in fixed-effects regressions represent the average value of the fixed effects. *t*-statistics are in parentheses.
* Significant at the 5 percent level.
** Significant at the 1 percent level.

Table 2: Race composition regressions

TABLE 3
REGRESSIONS OF NEWSPAPER SALES ON POPULATION COMPOSITION BY HISPANIC STATUS (N=11,584)
Dependent Variable: Zip Code per Capita Newspaper Sales, Hispanics and Non-Hispanics

	(1)	(2)	(3)	(4)	(5)
MSA Non-Hispanic population (α_1)	.0077 (1.73)	.0040 (1.08)	.0027 (.76)	.0034 (.91)	
MSA Hispanic population (α_2)	.0177 (.59)	.0395 (1.45)	.0459 (1.66)	.0345 (1.24)	
Zip Hispanic fraction	-.1054 (5.40)**	-.1089 (4.04)**	-.3916 (7.46)**	-.3668 (8.28)**	-.2893 (11.42)**
Zip Hispanic fraction $\times$ MSA non-Hispanic population ($\beta_1 - \alpha_1$)	-.0371 (3.56)**	-.0419 (4.11)**	-.0164 (1.17)	-.0330 (2.50)*	-.0395 (8.26)**
Zip Hispanic fraction $\times$ MSA Hispanic population ($\beta_2 - \alpha_2$)	-.0369 (.52)	-.0065 (.11)	-.1327 (1.48)	-.0572 (.59)	.2182 (5.55)**
β_1	-.029 (4.20)**	-.038 (4.48)**	-.014 (1.06)	-.030 (2.51)*	
β_2	-.019 (.38)	.033 (.72)	-.087 (1.13)	-.023 (.28)	
MSA fraction black				-.0544 (1.12)	
MSA fraction college educated				.3788 (3.51)**	
MSA fraction poor (income <\$15,000)				.0765 (1.09)	
MSA fraction old (age ≥ 65)				.4116 (3.81)**	
Census regions	no	yes	yes	yes	no
Interacted regions	no	no	yes	yes	no
MSA fixed effects	no	no	no	no	yes
Constant	.2037 (37.92)**	.1589 (8.97)**	.1610 (9.09)**	.0541 (1.10)	.2290 (190.15)**

NOTE.—All regressions are population weighted with standard errors clustered by MSA. Constants in fixed-effects regressions represent the average value of the fixed effects. *t*-statistics are in parentheses.
* Significant at the 5 percent level.
** Significant at the 1 percent level.

Table 3: Hispanic-status regressions

6.3 Table 4: Regressions by Education, Income, and Age and Table 6: MSA Fraction Hard News and MSA Demographic Composition

Read:

- Table 4 asks whether similar preference externalities appear for college education, poverty, and age 65+.
- The direct zip-code group shares show baseline readership differences: college-educated, nonpoor, and older zip codes tend to have higher readership.
- The interaction patterns are much weaker than for race and Hispanic status. Few own- and cross-population effects are statistically robust.
- This weak evidence is consistent with the paper's argument that education, income, and age are less clean preference groups and vary less across MSAs.
- Table 6 links product positioning to market composition. The dependent variable is the MSA share of reporters/editors assigned to hard news.
- The fraction black is negatively and significantly associated with the hard-news share. Other demographic variables are insignificant in the multivariate specification.
- This is consistent with Table 5's evidence that black zip codes are less likely to choose the broadsheet/hard-news-oriented paper in two-paper cities.

Economic message: Table 4 shows where the preference-externality test is weak; Table 6 helps explain where it is strong. Race predicts newspaper positioning in ways consistent with the circulation evidence.

TABLE 4
REGRESSIONS OF NEWSPAPER SALES ON POPULATION COMPOSITION BY EDUCATION,
INCOME, AND AGE ($N=11,584$)
Dependent Variable: Zip Code per Capita Newspaper Sales

	COLLEGE EDUCATED		POOR (Income ≤\$15,000)		OLD (Age ≥65)	
	(1)	(2)	(3)	(4)	(5)	(6)
MSA nongroup population (α_1)	-.0163 (1.79)		.0055 (.55)		.0290 (2.49)*	
MSA group population (α_2)	.0497 (1.05)		.0383 (.87)		-.1576 (1.70)	
Zip group fraction	.2191 (10.74)**	.2604 (12.95)**	-.1041 (4.41)**	-.1353 (5.76)**	.4939 (9.51)**	.5210 (10.29)**
Zip group fraction × MSA non-group population ($\beta_1 - \alpha_1$)	.0575 (2.91)**	.0412 (1.68)	-.0110 (.25)	.0066 (.13)	-.0956 (1.90)	-.1023 (1.72)
Zip group fraction × MSA group population ($\beta_2 - \alpha_2$)	-.1504 (1.44)	-.1187 (1.01)	-.2457 (1.36)	-.2812 (1.28)	.4359 (1.35)	.4360 (1.21)
β_1	.041 (2.82)**		-.006 (.14)		-.067 (1.43)	
β_2	-.101 (1.42)		-.207 (1.32)		.278 (.95)	
MSA fixed effects	no	yes	no	yes	no	yes
Constant	.1228 (6.97)**	.1433 (47.51)**	.2694 (21.72)**	.2605 (72.70)**	.0997 (5.40)**	.1537 (32.61)**

NOTE.—For economy of space, we report regressions along all three dimensions in the same table, referring to population categories and their complements as “group” and “nongroup.” All regressions are population weighted with standard errors clustered by MSA. Specifications in cols. 1, 3, and 5 include dummy variables for census regions. Constants in fixed-effects regressions represent the average value of the fixed effects. *t*-statistics are in parentheses.

* Significant at the 5 percent level.

** Significant at the 1 percent level.

Table 4: Education, income, and age regressions

TABLE 6
MSA FRACTION OF REPORTERS IN HARD NEWS AND MSA DEMOGRAPHIC COMPOSITION
($N=269$ MSAs)
Dependent Variable: MSA Fraction Hard News

	(1)	(2)	(3)	(4)	(5)	(6)
MSA fraction black	-.191 (3.16)**					-.210 (2.82)**
MSA fraction Hispanic		.027 (.68)				-.028 (.61)
MSA fraction college educated			.121 (1.19)			.086 (.73)
MSA fraction poor (income ≤\$15,000)				.117 (.96)		-.023 (.16)
MSA fraction old (age ≥65)					-.052 (.29)	-.032 (.16)
MSA population (M)	.012 (1.98)*	.010 (1.59)	.007 (1.16)	.012 (1.76)	.009 (1.48)	.010 (1.42)
Constant	.597 (67.25)**	.557 (16.51)**	.557 (27.54)**	.554 (20.26)**	.586 (24.89)**	.615 (9.87)**

NOTE.—*t*-statistics are in parentheses.

* Significant at the 5 percent level.

** Significant at the 1 percent level.

Table 6: Hard-news share and demographics

6.4 Table 5: Newspaper Market Shares and Zip Code Demographic Composition

Read:

- Table 5 uses cities with two major daily newspapers, usually an upscale broadsheet and a tabloid competitor.
- The dependent variable is the broadsheet’s market share among the two major papers in a city.
- Broadsheet share is lower in zip codes with larger black and Hispanic shares.
- Broadsheet share is higher in zip codes with larger college-educated shares and lower poverty shares.
- The race coefficients remain large after controlling for other demographics, suggesting that race captures preference differences not fully proxied by income or education.
- The result supports the idea that different demographic groups prefer different bundles of newspaper content.

Economic message: This table provides the mechanism evidence. If different groups systematically choose different newspaper types when both are available, then firms have an incentive to position newspapers differently when market demographic composition changes.

TABLE 5
NEWSPAPER MARKET SHARES AND ZIP CODE DEMOGRAPHIC COMPOSITION
Dependent Variable: Newspaper Market Share

	CHICAGO TRIBUNE (CHICAGO SUN- TIMES)		PHILADELPHIA INQUIRER (PHILADELPHIA DAILY NEWS)		BOSTON GLOBE (BOSTON HERALD)		WASHINGTON POST (WASHINGTON TIMES)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Zip fraction black	-.449 (14.94)**	-.365 (11.40)**	-.476 (12.46)**	-.13 (2.77)**	-.401 (5.55)**	-.132 (2.53)*	-.099 (7.52)**	.028 -1.46
Zip fraction Hispanic	-.527 (6.96)**	-.372 (6.98)**	-.916 (5.48)**	.163 -1.11	-.525 (3.99)**	.641 (5.60)**	.311 (3.70)**	.294 (4.28)**
Zip fraction college educated	.963 (18.48)**	.576 (13.55)**	.726 (8.82)**	.125 -1.68	.674 (14.51)**	.497 (12.10)**	.162 (7.15)**	.082 (3.02)**
Zip fraction poor (income ≤\$15,000)	-1.221 (17.04)**	.012 -1.14	-1.318 (18.00)**	-1.100 (8.05)**	-1.189 (12.19)**	-1.086 (9.03)**	-.500 (10.03)**	-.489 (6.57)**
Zip fraction old (age ≥65)	-.241 -.89	-.785 (5.83)**	.181 -.62	.379 -1.78	-.748 (2.85)**	-.214 -1.24	-.190 (2.07)*	-.024 -.29
Observations (zips)	282	282	259	259	244	244	184	184

NOTE.—Dependent variable is the market share of the top among both listed papers (e.g., *Chicago Tribune*) by zip code in the home market (MSA). The first column for each paper reports coefficient estimates for five univariate regressions. The second column reports estimates for a multivariate regression. *t*-statistics are in parentheses.

* Significant at the 5 percent level.

Table 5: Newspaper Market Shares and Zip Code Demographic Composition

7 Short summary

- The paper studies preference externalities in daily newspaper markets.
- Preference externalities arise when firms tailor products toward larger consumer groups, making those products more appealing to members of that group.
- The core data combine zip-code circulation, zip-code demographics, MSA demographics, and newspaper content/staffing measures.
- The main empirical challenge is that sales by race are not observed; the paper uses zip-code demographic shares to infer group-specific readership gaps.
- Blacks are more likely to buy newspapers in markets with larger black populations and less likely in markets with larger white populations.
- Whites are more likely to buy newspapers in markets with larger white populations and are mostly unaffected by black population.
- Similar but weaker evidence appears for Hispanics and non-Hispanics.
- There is little robust evidence for education, income, or age groups.
- Product positioning evidence supports the mechanism: different groups prefer different newspapers in two-paper cities, and market content varies with demographic composition.
- The paper contributes to differentiated-products research by showing that the supply of product characteristics varies with the composition of consumers.

8 Conclusion

8.1 Core conclusion

George and Waldfogel show that market composition affects consumption through product supply and positioning. In daily newspaper markets, consumers are more likely to buy when there are more local consumers with similar preferences. This is especially clear for race and, to a lesser extent, Hispanic status.

8.2 Incremental contribution relative to prior work

The paper adds an empirical supply-side dimension to differentiated-products economics. Standard demand estimation asks how consumers choose among available products; this paper asks how the available products themselves depend on who lives in the market. It is therefore a bridge between media economics, local public-good-like product provision, and differentiated-products industrial organization.

8.3 Mechanism and interpretation

The evidence is consistent with a Hotelling/Spence logic. Fixed costs limit the number of newspapers or the extent of content adaptation. Firms therefore target products toward large groups. When a group is small, its members face products farther from their ideal points and buy less often. When the group becomes larger, firms shift content or introduce/maintain more appealing products, increasing within-group consumption.

8.4 Welfare interpretation

The paper cannot formally measure consumer surplus. Still, the results suggest that same-group consumers confer positive externalities on each other. If prices do not rise and quality does not fall, additional black consumers appear to benefit other black newspaper consumers by inducing better targeted newspapers. Similar logic applies to white consumers. Cross-group effects can be negative when targeting one group makes the product less attractive to another.

8.5 Policy and broader relevance

The results matter beyond newspapers. Any market with fixed costs, heterogeneous tastes, and limited product variety can generate similar preference externalities. Examples include local media, restaurants, retail, cultural goods, health services, and digital platforms with local network effects. The paper also informs debates about minority media access: underrepresentation in markets can lead not only to lower consumption but to lower supply of preferred product attributes.

8.6 Limitations and open questions

The evidence is not experimental. MSA demographic composition may correlate with unobserved local newspaper demand or supply conditions. The paper also observes circulation and content categories but not full content quality or individual purchases. Future research could estimate consumer surplus directly, exploit exogenous demographic shocks, or study how digital media changes fixed costs and therefore the strength of preference externalities.

8.7 Takeaway

The central takeaway is that consumer groups do not merely compete for existing products; they shape what products exist and how products are positioned. In markets with fixed costs and differentiated tastes, who else lives in the market changes what each consumer can buy.